

■ README.md — Complete Content Guide

Scuba Diving Underwater Health Monitoring System using Li-Fi Technology

Section 1 — Project Title & Badges

- # Scuba Diving Underwater Health Monitoring System using Li-Fi Technology
- Add shields.io badges: ESP32 | Li-Fi | Arduino IDE | IoT | License: MIT
- Department: ECE — CITNC | Year: 2025–2026

Section 2 — Project Description

- What it does: Real-time health monitoring of scuba divers using Li-Fi communication
- Why it matters: Traditional RF/Wi-Fi fail underwater; Li-Fi uses light for reliable transmission
- Key innovation: Air balloon emergency ascent + surface alert system
- Who benefits: Scuba divers, marine researchers, naval divers, rescue teams

Section 3 — Problem Statement

- Scuba divers face risks due to lack of effective underwater communication
- RF and Wi-Fi signals fail underwater — critical health data cannot be transmitted
- Delayed emergency response increases risk of injury or death
- This project solves it using Li-Fi (light-based) communication

Section 4 — Objectives

- Real-Time Health Monitoring: Heart rate, SpO2, body temperature
- Reliable Underwater Communication: Li-Fi light-based data transmission
- Abnormal Health Detection: Algorithm triggers immediate alerts
- Emergency Ascent System: Air balloon auto-inflation mechanism
- Surface Alert System: Real-time notifications to monitoring team

Section 5 — System Architecture

- Transmitter Module (Diver Unit): ESP32 + Heartbeat sensor + Pulse oximeter + Temperature sensor + Panic switch + Li-Fi Tx
- Receiver Module (Surface Team): Li-Fi Rx + ESP32 + LCD Display + Buzzer + Cloud
- Li-Fi Communication Link: LED → Photodiode (underwater optical data transfer)
- Add block diagram images: transmitter_block.png, receiver_block.png

Section 6 — Components Used

- ESP32 Microcontroller (Dual-core, Wi-Fi + Bluetooth)
- MAX30102 Pulse Oximeter + Heartbeat Sensor
- Waterproof Temperature Sensor (DS18B20)

- Li-Fi Transmitter (LED + LM741 pre-amp + power amp)
- Li-Fi Receiver (Photodiode/LDR + LM358 amp)
- LCD Display (16x2), Relay Module, Air Motor (Pneumatic)
- Buzzer, Panic Switch (Waterproof push button)
- Power Supply: 7805 & LM317 voltage regulators

Section 7 — Working Principle

- Sensors on diver continuously measure heart rate, SpO2, temperature
- ESP32 (Transmitter) processes data and encodes it as Li-Fi light signals via LED
- Photodiode (Receiver) captures light, decodes data via ESP32
- LCD displays diver vitals on surface; buzzer alerts for abnormal readings
- If vitals cross threshold: Relay activates → Air motor inflates balloon → Diver ascends
- Panic switch allows manual SOS signal from diver

Section 8 — Pin Connections Table

- ESP32 to MAX30102 (I2C: SDA, SCL)
- ESP32 to Temperature Sensor (One-wire: GPIO4)
- ESP32 to Li-Fi Transmitter LED (GPIO pin via transistor)
- ESP32 to Relay (GPIO output)
- ESP32 to Buzzer (GPIO output)
- ESP32 to Panic Switch (GPIO input with pull-up)
- Li-Fi Receiver → ESP32 ADC/UART input

Section 9 — Setup & Installation

- Step 1: Install Arduino IDE
- Step 2: Add ESP32 board support URL in preferences
- Step 3: Install required libraries: MAX30102, OneWire, DallasTemperature, LiquidCrystal_I2C
- Step 4: Clone this repository
- Step 5: Upload transmitter.ino to Diver ESP32
- Step 6: Upload receiver.ino to Surface ESP32
- Step 7: Open Serial Monitor (115200 baud) and test

Section 10 — Results & Output

- Heart rate graph (ThingSpeak): normal range 72–110 BPM, accuracy ± 2 –3 BPM
- Temperature graph: smooth transitions, detected hypothermia early
- SpO2 graph: accuracy $\pm 2\%$, responds instantly to low oxygen
- Li-Fi latency: <100 ms, near-zero packet loss
- Buzzer response: 0.5 sec | Relay response: 1.2 sec | Air motor: 1.8 sec
- Add screenshots: heartbeat_graph.png, temperature_graph.png, spo2_graph.png

Section 11 — Limitations

- Li-Fi range limited to 2–3 meters underwater (line-of-sight required)
- Sensitive to water turbidity, suspended particles, and ambient light
- High LED power consumption reduces battery life
- Currently supports single diver only
- Sensor accuracy may drift due to wet skin or loose placement

Section 12 — Future Scope

- Laser-based Li-Fi for extended underwater range
- Hybrid acoustic + optical communication for turbid water
- AI-based predictive health alerts
- Multi-diver simultaneous monitoring
- IP68 waterproof housing for all components
- Depth, breathing rate, and orientation sensors

Section 13 — References

- List all 11 references from the project report
- IEEE, Oceans, Optics Communications, Journal of Marine Science

Section 14 — License

- MIT License recommended for academic/open-source projects